

Could they have seen their waves coming?

COVID-19 analytics on the Johns Hopkins time series — regression, clustering and graph analytics turned into an early-warning recommendation for the focal country's neighbours.

103.8M

US confirmed cases — worst outbreak in the dataset

164

weeks of data · Jan 2020 – Mar 2023

3

methods chained: regression → clustering → graph

THE APPROACH

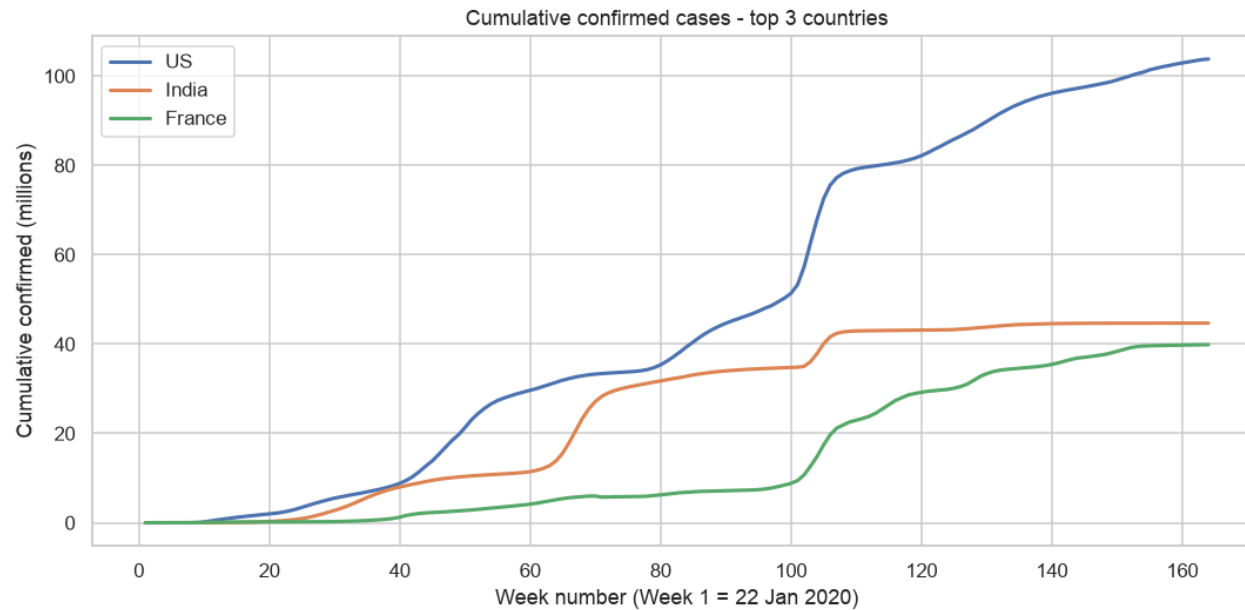
One connected chain: from raw counts to a decision



Engine: Apache Spark MLlib (regression + K-Means) · networkx (graph) · 164 weeks of Johns Hopkins confirmed-case data

Narrating the decision, not the code — every step exists to answer one question for the neighbours.

The three worst-hit countries: same climb, different shapes

**103.8M**

United States

44.7M

India

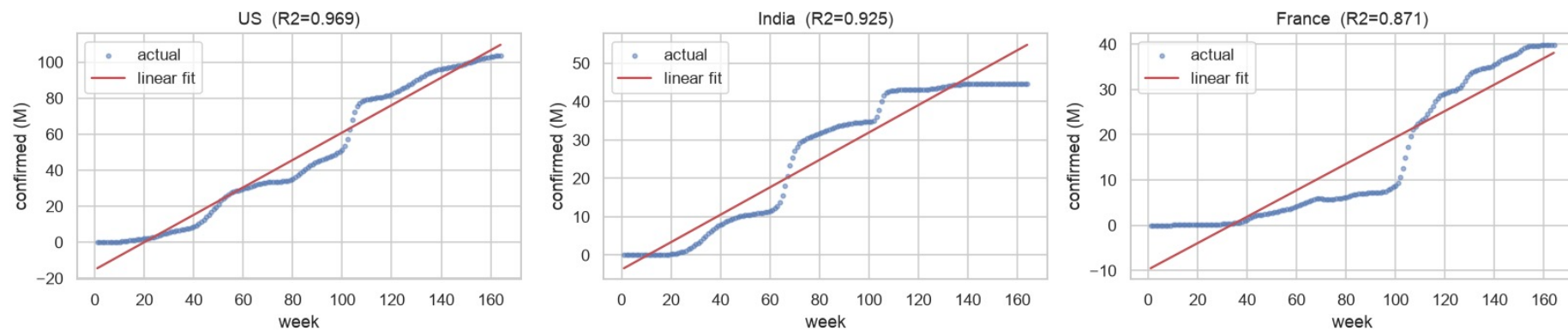
39.9M

France

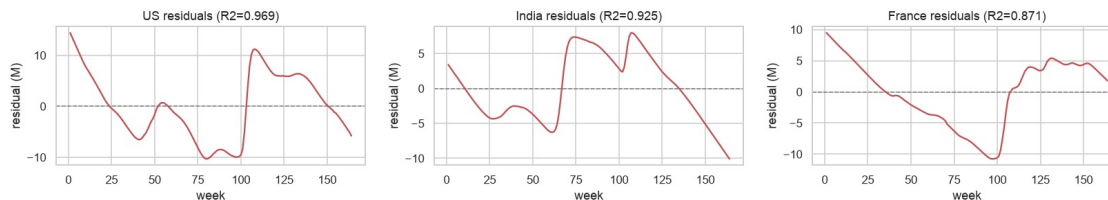
Which of the three is the most volatile — the one whose surges are sharpest?

The US is the most volatile, but $R^2 = 0.97$ hides the story

Linear regression fits - cumulative confirmed vs week



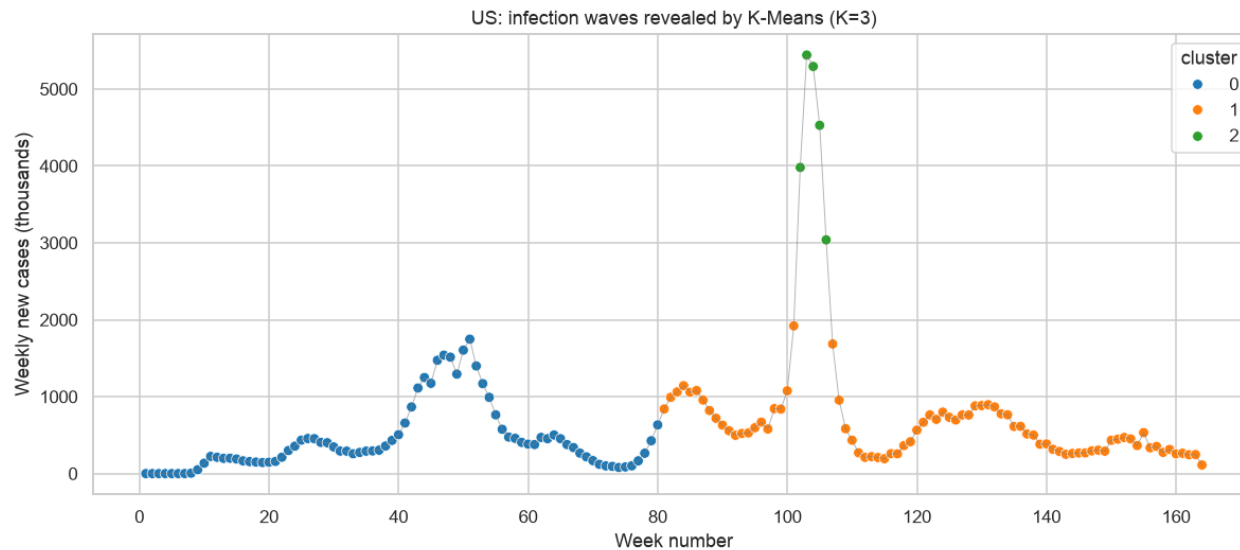
Residuals are S-shaped, not noise - the straight line is systematically wrong



Volatility = variance of weekly new cases:
US 6.4×10^{11} vs India 2.5 · France 1.6.

Slope $\approx$ 760k cases/week, $R^2 = 0.97$ - yet the S-shaped residuals show the straight line is systematically wrong. It can't say when surges hit.

Clustering reveals what the line hid: growth came in waves



K = 3

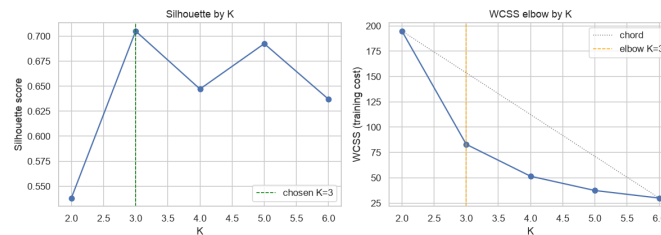
silhouette 0.705 — clean separation, chosen by iteration K = 2...6

4.46M/wk

the isolated mega-surge — weeks 102–106, January 2022, Omicron

≈ 7×

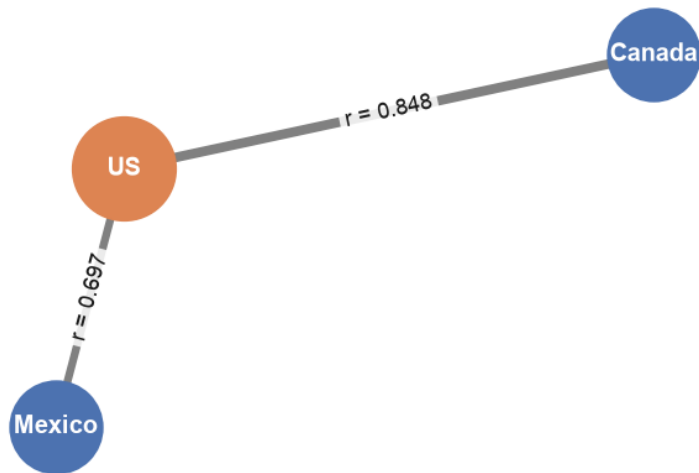
that one cluster vs the overall weekly average



*“Cases went up”
becomes
“Here is exactly when, and how hard.”*

The obvious reading says Canada: and it answers the wrong question

US and two neighbours that do not border each other
(edge = correlation of weekly new cases)



$r = 0.85$

Canada — weekly new cases move almost in lockstep with the US

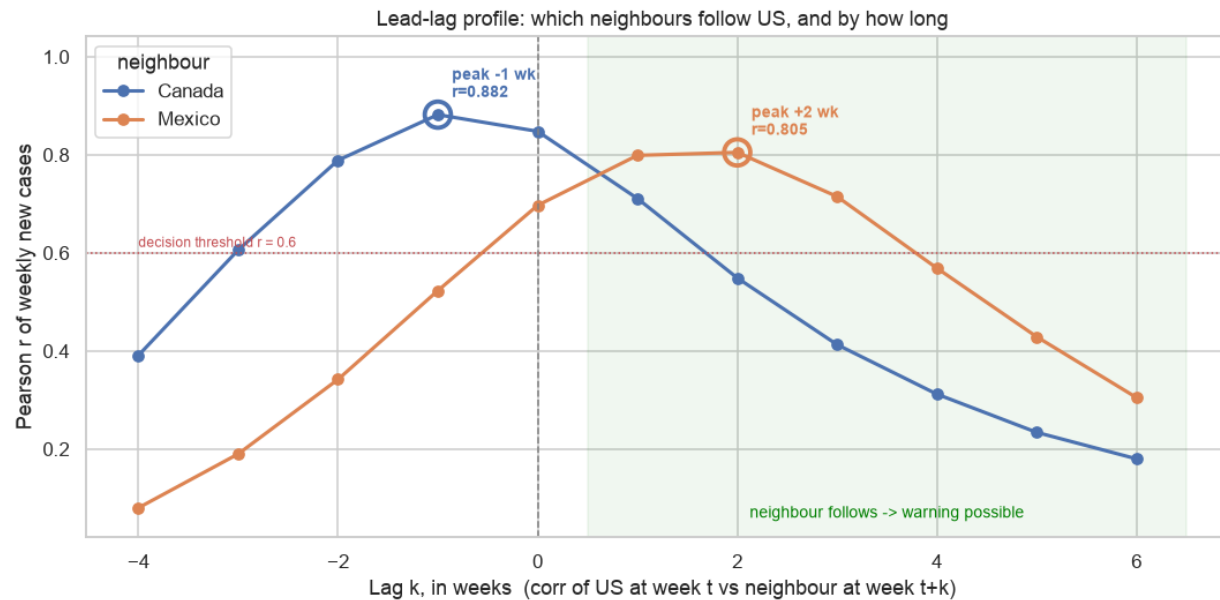
$r = 0.70$

Mexico — weaker same-week correlation

Same-week correlation asks whether two curves look alike.

What a neighbour needs to know is whether its wave arrives after the American one — if both peak in the same week, there is nothing to warn about.

Shift the series... and the ranking reverses



Decision rule (fixed before looking):

warn only if correlation peaks at lag $\geq +1$ week AND $r \geq 0.6$ at that lag.

Canada

peaks at -1 week ($r = 0.88$) — moves with, or ahead of, the US. No warning window.

Mexico

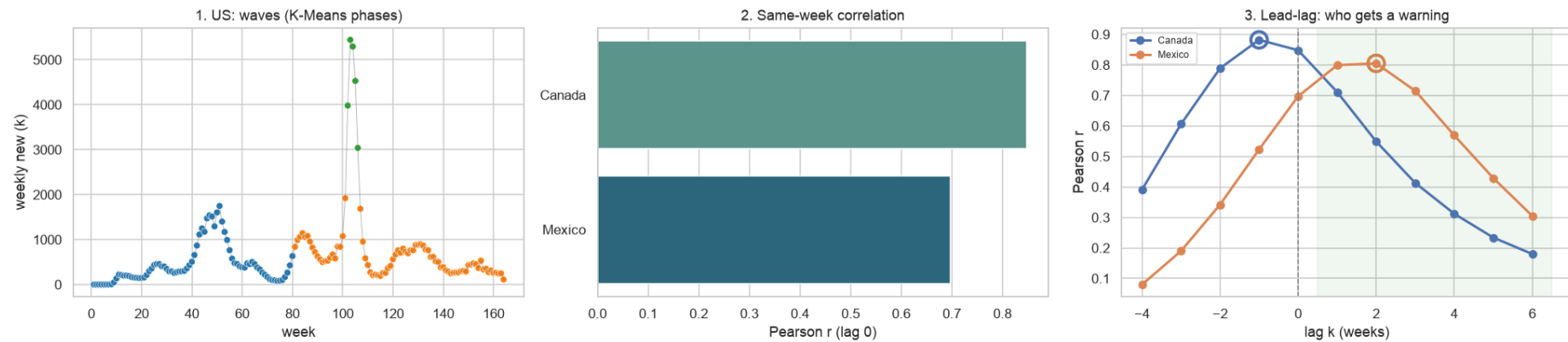
climbs from 0.70 to $r = 0.81$ at +2 weeks — genuinely follows the US by a fortnight.

The neighbour that looked weaker is the one that can actually act on the signal.

THE WHOLE STORY

Raw cases → the phases clustering found → who they can warn

From raw cases to a neighbour early-warning recommendation



Read left to right: the US waves coloured by cluster phase; the same-week correlation that pointed at Canada; the lead-lag profile that overturns it.

What the neighbours should actually do

Mexico: The real early-warning case



$r = 0.81$ at a two-week lag. When the US enters a surge phase, pre-position testing and hospital capacity ~2 weeks ahead of the expected peak.

Same-week correlation alone would have missed this.

Canada: Invest domestically



Highly correlated but synchronous: by the time US numbers climb, Canadian ones already are.

The US is not a leading indicator - Canada's money belongs in its own surveillance.

Everyone: Plan for the peak, not the average



Capacity must be sized for the Omicron-style mega-surge cluster (~7× the weekly average).

Planning for the average is how you get overwhelmed by the peak.

HONEST CAVEATS · AND THE RESULT THAT STANDS

Limitations — and the core result

- Confirmed cases measure testing and reporting as much as transmission; the file ends 9 March 2023 - JHU stopped collecting the next day.
- Week number was a clustering input, so phases are partly temporal by construction.
- “Neighbour” means geography, not population mobility; one lag was fitted across three years while variants travelled at different speeds.
- Correlation is not causation - a lagged match may mean a variant reached both countries in sequence.
- Next steps: mobility and vaccination data; a lag estimated per wave rather than once.

The same-week number said

Canada.

Shifting the series two weeks said

Mexico.

That is the recommendation that would actually have bought someone time.